

1. Consider a vanilla call option on an underlying asset S with expiration date T and strike K under the Black–Scholes–Merton model. Let $\tau = T - t$ be the time to maturity and denote by Γ the gamma of the option.

The Black–Scholes formula for a call is

$$C(S, t) = SN(d_1) - Ke^{-r\tau}N(d_2),$$

where

$$d_{1,2} = \frac{\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad \varphi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}, \quad N'(x) = \varphi(x).$$

The gamma is

$$\Gamma(S, t) = \frac{\partial^2 C}{\partial S^2} = \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}}.$$

(a) We maximize Γ with respect to S at fixed t (hence fixed τ). Since $\Gamma > 0$, we can maximize $\ln \Gamma$:

$$\ln \Gamma = \ln \varphi(d_1) - \ln S - \ln(\sigma\sqrt{\tau}).$$

Using $\ln \varphi(d_1) = \text{const} - \frac{1}{2}d_1^2$ and

$$d_1(S) = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad \frac{\partial d_1}{\partial S} = \frac{1}{S\sigma\sqrt{\tau}},$$

we obtain

$$\frac{\partial}{\partial S} \ln \Gamma = -d_1 \frac{\partial d_1}{\partial S} - \frac{1}{S} = -\frac{d_1}{S\sigma\sqrt{\tau}} - \frac{1}{S}.$$

Setting this derivative to zero gives

$$-\frac{d_1}{S\sigma\sqrt{\tau}} - \frac{1}{S} = 0 \implies d_1 = -\sigma\sqrt{\tau}.$$

Substituting the definition of d_1 ,

$$\frac{\ln(S^*/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}} = -\sigma\sqrt{\tau} \implies \ln \frac{S^*}{K} = -(r + \frac{3}{2}\sigma^2)\tau.$$

Hence

$$\boxed{S^*(t) = K \exp(-(r + \frac{3}{2}\sigma^2)(T - t))}$$

is the stock price at which the gamma is maximized at time t . Since $\Gamma(S, t) \rightarrow 0$ as $S \rightarrow 0$ and as $S \rightarrow \infty$, and there is a single critical point, this critical point is a global maximum.

(b) Now assume that at time t the spot equals the strike, $S = K$, and consider Γ as a function of τ :

$$d_1(\tau) = \frac{\ln(1) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}} = \frac{(r + \frac{1}{2}\sigma^2)\sqrt{\tau}}{\sigma}, \quad \Gamma(\tau) = \frac{\varphi(d_1(\tau))}{K\sigma\sqrt{\tau}}.$$

Again maximize $\ln \Gamma$:

$$\ln \Gamma(\tau) = \ln \varphi(d_1(\tau)) - \ln K - \ln \sigma - \frac{1}{2} \ln \tau.$$

Write $A = \frac{r + \frac{1}{2}\sigma^2}{\sigma} > 0$, so $d_1(\tau) = A\sqrt{\tau}$. Then

$$\frac{d}{d\tau} \ln \Gamma(\tau) = -d_1(\tau) \frac{dd_1}{d\tau} - \frac{1}{2\tau} = -A\sqrt{\tau} \cdot \frac{A}{2\sqrt{\tau}} - \frac{1}{2\tau} = -\frac{A^2}{2} - \frac{1}{2\tau} < 0 \quad (\tau > 0).$$

Thus $\ln \Gamma(\tau)$ (and hence $\Gamma(\tau)$) is strictly decreasing in τ . Therefore Γ is maximized for the smallest possible τ , i.e. as $\tau \rightarrow 0^+$, which corresponds to $t \rightarrow T^-$. In the limit

$$\Gamma(\tau) \sim \frac{1}{K\sigma\sqrt{2\pi}\tau} \xrightarrow{\tau \rightarrow 0^+} \infty.$$

Hence, when $S = K$, the gamma becomes arbitrarily large as we approach maturity:

$$\boxed{\Gamma \text{ reaches its maximum in the limit } t \rightarrow T^-}.$$

2. Let $x \in \mathbb{R}^n$, $z \in \mathbb{R}^m$, $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$, $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and

$$h(x) = f(x, g(x)).$$

Write

$$f_x = \frac{\partial f}{\partial x} \in \mathbb{R}^n, \quad f_z = \frac{\partial f}{\partial z} \in \mathbb{R}^m, \quad J_g = \frac{\partial g}{\partial x} \in \mathbb{R}^{m \times n},$$

and denote by f_{xx}, f_{xz}, f_{zz} the corresponding Hessian blocks.

First derivative. By the chain rule,

$$\nabla h(x) = f_x(x, g(x)) + J_g(x)^\top f_z(x, g(x)).$$

Second derivative. Differentiate again with respect to x :

$$\begin{aligned} \nabla^2 h(x) &= \frac{\partial}{\partial x} \left[f_x(x, g(x)) \right] + \frac{\partial}{\partial x} \left[J_g(x)^\top f_z(x, g(x)) \right] \\ &= f_{xx}(x, g(x)) + f_{xz}(x, g(x)) J_g(x) \\ &\quad + J_g(x)^\top f_{zx}(x, g(x)) + J_g(x)^\top f_{zz}(x, g(x)) J_g(x) \\ &\quad + \sum_{k=1}^m f_{z_k}(x, g(x)) \nabla^2 g_k(x). \end{aligned}$$

In index notation, for $i, j = 1, \dots, n$,

$$\begin{aligned} \frac{\partial^2 h}{\partial x_i \partial x_j} &= \frac{\partial^2 f}{\partial x_i \partial x_j} + \sum_{k=1}^m \frac{\partial^2 f}{\partial x_i \partial z_k} \frac{\partial g_k}{\partial x_j} + \sum_{k=1}^m \frac{\partial^2 f}{\partial x_j \partial z_k} \frac{\partial g_k}{\partial x_i} \\ &\quad + \sum_{k, \ell=1}^m \frac{\partial^2 f}{\partial z_k \partial z_\ell} \frac{\partial g_k}{\partial x_i} \frac{\partial g_\ell}{\partial x_j} + \sum_{k=1}^m \frac{\partial f}{\partial z_k} \frac{\partial^2 g_k}{\partial x_i \partial x_j}. \end{aligned}$$

This shows that, in general,

$$\frac{\partial^2 h}{\partial x^2} \neq \frac{\partial^2 f}{\partial x^2},$$

because there are additional contributions involving derivatives of g and the mixed and z -derivatives of f .

To obtain equality, we must include all these extra terms:

$$\boxed{\frac{\partial^2 h}{\partial x^2} = \frac{\partial^2 f}{\partial x^2} + f_{xz} J_g + J_g^\top f_{zx} + J_g^\top f_{zz} J_g + \sum_{k=1}^m f_{z_k} \nabla^2 g_k}$$

(or equivalently the indexed form above). Only if g is constant (so $J_g = 0$ and $\nabla^2 g_k = 0$) and f does not depend on z do these additional terms vanish, in which case $\partial^2 h / \partial x^2 = \partial^2 f / \partial x^2$.